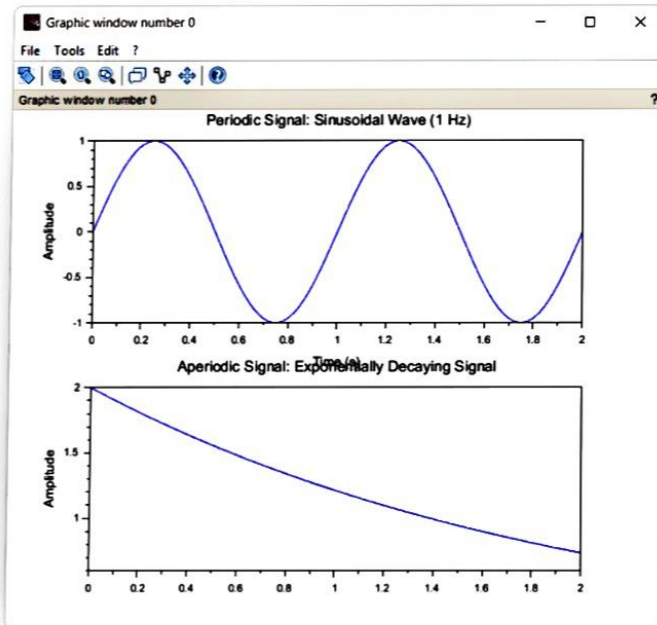


```
exp 2.1.sci (C:\Users\imjan\OneDrive\Documents\exp 2.1.sci) - SciNotes
File Edit Format Options Window Execute ?

1.1 exp.sci 1.2 exp.sci exp 2.1.sci 2.2.sci

1 // Test Case: Generation of Periodic and Aperiodic Signals
2 // Parameters for the periodic signal (Sinusoidal)
3 f_periodic = 1; // Frequency in Hz
4 A_periodic = 1; // Amplitude
5 t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds
6 // Generate the periodic signal
7 x_periodic = A_periodic * sin(2 * pi * f_periodic * t_periodic);
8 // Plot the periodic signal
9 clf; // Clear the current figure
10 subplot(2,1,1); // Subplot 1 for the periodic signal
11 plot(t_periodic, x_periodic);
12 xlabel('Time (s)');
13 ylabel('Amplitude');
14 title('Periodic Signal: Sinusoidal Wave (1 Hz)');
15 // Parameters for the aperiodic signal (Exponentially Decaying)
16 a_aperiodic = 0.5; // Base of the exponential decay
17 A_aperiodic = 2; // Amplitude
18 t_aperiodic = 0:0.01:2; // Time vector for 0 to 2 seconds
19 // Generate the aperiodic signal
20 x_aperiodic = A_aperiodic * exp(-a_aperiodic * t_aperiodic);
21 // Plot the aperiodic signal
22 subplot(2,1,2); // Subplot 2 for the aperiodic signal
23 plot(t_aperiodic, x_aperiodic);
24 xlabel('Time (s)');
25 ylabel('Amplitude');
26 title('Aperiodic Signal: Exponentially Decaying Signal');
```



```

11 t = -1:0.001:1;
2 x = exp(-abs(t)).*sin(4*pi*t) + 0.3; // Reverse signal
3 x_rev = interp(t, x, -t, 'linear'); // Even and odd parts
4 xe = 0.5 * (x + x_rev);
5 xo = 0.5 * (x - x_rev);
6 clf; plot(t, x, 'k');
7 plot(t, xe, 'b--');
8 plot(t, xo, 'r:');
9 legend("x(t)", "Even Part", "Odd Part");
10 xtitle("Even-Odd Decomposition", "t", "Amplitude");
11

```

